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Advanced information management for process sciences: knowledge-based documentation of mathematical models

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Abstract: The development of mathematical models is a basic as well as a knowledge-intensive activity of the engineering disciplines and applied sciences. The modelling process involves actions and data that have great influence on the way the implementation of a model can be applied for a particular task, reused (on a component base) or extended. However, much of the corresponding information gets lost. In the area of scientific work, publications and reports are available as complementary information sources for models together with data sources resulting from experimental work used for model development. Concerning models for physico-chemical processes, this information base is, in practice, often not sufficient and cannot be retrieved and accessed in a convenient way. At this point, the well-known methodology of ontology-based information integration provides a promising approach. The basic idea and design principles of ontologies as common data models as well as the idea of information integration is explained. A concept for the enrichment and integration of different information resources concerning mathematical models is described, which combines the documentation of models and the management of related information resources.

Keywords: information integration; information management; mathematical models; modelling process; ontologies.

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Professor Dr Günter Wozny obtained his chemical process engineer grade in 1976 from the RWTH Aachen, Germany. He finished his PhD thesis in the field of thermodynamics and phase equilibria in 1979 at the RWTH Aachen. He wrote his habilitation thesis in the area of process simulation and energy optimisation at the University of Siegen, Germany in 1983. From 1983 to 1993, he was Chief of the Department of Process and Plant Technology at the Henkel Co., in Dusseldorf. In 1993 he became full Professor for process dynamics and operation at the Technical University of Berlin. Professor Wozny has authored more than 200 publications in the field of thermodynamics, simulation, control and optimisation.

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1 Introduction

Access to scientific information has always been crucial for research and technical development. This is quite obvious when it comes to the definition of research objectives. Research projects usually require a careful inquiry of existing work to ensure the uniqueness of new approaches. At this point, the possibilities and the efficiency of information retrieval play a major role.

Many database applications, digital libraries and search engines focus on aspects of retrieval such as indexing techniques and ranking for keyword based searches [1]. Those applications are mainly aimed at users in search of basic information, giving them an overview of the results of existing projects. Despite modern retrieval techniques, the information itself is in most cases, represented and transferred in the traditional, document-based way – in the form of articles, reports, presentations and spreadsheets.

However, this might not be sufficient when scientific resources serve as a base for new approaches or applications built on existing results. Scientific resources in this form provide limited support for modular access [2] and precise retrieval. This aspect as well as the problem of proprietary data formats in terms of long-term archiving has been discussed previously [3,4]. In addition, most document-based approaches ignore connections between different, possibly heterogeneous resources. An example of the latter would be a scientific article, which is based on results in the form of measured data. Even though the article itself might provide an adequate amount of information about research objectives and the interpreted results, it does not contain enough information about the database, i.e. in which way data was obtained, how it was filtered and processed, etc. This is especially evident in the area of process modelling, which is described in Section 3.

Text-based information sources about process models usually contain a number of elements, such as model parameters, that are derived out of existing data, either directly or as a result of data analysis. Furthermore most process models build on existing work in terms of composition or extension of available models, which leads to additional semantic connections to other resources.

Since the main purpose of an article or a report is not to serve as a kind of manual or user guide, it does not give the necessary information to help comprehend the entire modelling process that led to the results described in the particular publication. While this

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is not necessary in terms of the general scientific communication process (awareness of new outcomes, ownership protection etc. [5], it may not be sufficient concerning the extension and reuse of existing research results. The next section explains how the value of information resources is extended by data models (or, more precisely, ontologies) helping to describe semantic connections and to provide an integrated access.

2 The concept of ontologies for knowledge-based applications and distributed (web-based) resources

Semantic techniques have recently gained much attention, especially in the area of internet applications. While the basic concept of ontologies has been known for some time, it provides an approach to overcoming problems concerning the retrieval techniques currently available for the internet [6].

2.1 The definition of ontologies

A common interpretation of the term 'ontology' as it is used in the computer science community is given by Gruber, who specifies it as 'an explicit specification of a conceptualisation', while a 'conceptualisation' is described as 'an abstract simplified view of the world that we wish to represent for some purpose' [7]. The simplified view is represented on the base of a formal vocabulary, which is often (but not necessarily) part of a logical language [8]. In contrast to data models tailored to a specific application, an ontology emphasises communication and inter-operability [9], thus implying 'a shared and common understanding of some domain that can be communicated across people and computers' [10]. The notion of a common understanding is important for the view of ontologies in contrast to other approaches. One of the basic ideas behind the concept of ontologies is reuse of existing knowledge bases [11–14] for which a certain degree of generality is a key requirement. A perfect example is given by Uschold *et al*. [12] and as by Gruber [15] who have each developed an ontology for physical measures.

Guarino distinguishes between ontologies and knowledge bases by characterising ontologies as 'a *particular* knowledge base, describing facts assumed always to be true by a community of users, by virtue of the agreed-upon meaning of the vocabulary used' [8]. Guarino refines Gruber's definition and draws a precise distinction of conceptualisation and ontology, thus emphasising that conceptualisations are language-independent while ontologies are language-dependent. He shows that the adoption of a common vocabulary is not necessarily implying exchangeability of information (due to differing conceptualisations). As a consequence, following a top-down-approach may often be preferable for the development of ontology-driven information systems [8] (cp. also [16]), which does not exclude bottom-up approaches in general.

Several authors make out different categories of ontologies, such as domain and method ontologies [11] as well as 'application ontologies' [8]. Domain ontologies refer to certain domains (such as mechanical engineering) while task ontologies refer to the processing of a set of tasks (such as planning or scheduling) by applying suitable problem-solving methods [11,17,18].

2.2 Development of ontologies

Despite the definition of ontologies given above, it is in many cases still arguable, whether a knowledge base or parts of it could be termed ontology. Concerning the design of information systems and knowledge-based systems, the design principles developed by Gruber [15] are helpful in order to benefit from the concept of ontologies and not just used it as a new term for application specific data models. Gruber establishes the following design principles:

- clarity, i.e. minimisation of ambiguity by axiomatisation
- coherence, i.e. excluding definitions out of which facts could be inferred, that are incosistent with the underlying conceptualisation
- minimal encoding bias, i.e. leaving apart details of implementation concerning particular representation systems
- extendibility, i.e. providing a way to extend the ontology
- minimal ontological commitment, which in a sense complies with Gruber's suggestion of starting with a small top-level ontology.

Minimal ontological commitment allows for different systems to commit to a core ontology without necessarily committing to derived ontologies. The core ontology makes it possible for a system to 'understand' a different standard (e.g. perform unit conversions based on an ontology for physical measures – c.p. [15]).

The example of physical measures illustrated by Gruber [15] turns out to be an 'ideal' ontology, which cannot be expected to be developed for other domains in the same way. Reuse and sharing of knowledge bases is therefore not easy to establish. The development of an ontology is among other aspects (such as the representation language), influenced by the inferences intended to perform, which limits reusability and exchangeability [19]. In cases where reusability can be expected to be very limited, Guarino sees the ontology concept as an analogy to existing CASE (Computer Aided Software Engineering)-tools, supporting the analysis process for the design of information systems [8]. In this case, a top-down approach with a relatively small top-level ontology is preferable.

2.3 Applications of ontologies

There are various applications of ontologies, two of which are listed here (cp. [20]):

1. knowledge-based systems designed for typical tasks, such as configuration, planning and diagnosis (e.g. [21–23])
2. information systems (cp. [8]), where ontologies are used for the purpose of information integration (for an overview, cp. [24,25])
3. user interfaces that are based on ontologies and are able to reflect changes in the model at runtime (cp. [8]).

There are some more applications that are not listed here, like natural language processing (also in connection with information retrieval, such as in OntoSeek [26]) and standardisation for the purpose of data exchange or web-based retrieval [20].

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In many cases the last two applications are complementary. Integrating different information sources is a promising approach in cases where sources are distributed and/or of a heterogeneous type (unstructured text sources, semi-structured text sources, databases). The basic idea of the integration approach is to simply provide a common data model for different information sources, in order to simplify retrieval and access. In the case of heterogeneous sources, users interested in particular information are confronted with several problems, such as how to retrieve, access and interpret that information (cp. [27]). Searching through an information source such as a database for example requires knowledge about searching techniques (e.g. the query language) and the structure of that source (e.g. table structure). The problem is, that heterogeneity occurs at different levels: the syntactical, the structural and the semantic level [24,27]. The SEAL-system developed by Maedche *et al*. [28] is an example of an ontology-based integration platform, that combines information integration and website management. A common data model in combination with different forms of web-based user interfaces allows the retrieval information from different heterogeneous sources. The interconnection between model and sources is based on wrappers, which enrich the original source with metainformation.

The concept of information integration is important for the approach presented in this paper. It implies the *formalization* of domain knowledge and allows *semantic* information retrieval and navigation. The next section illustrates, that concerning the development of mathematical models for physico-chemical processes, not only the result in form of the implemented model but the modelling process should be considered in terms of knowledge sharing. This involves the creation of an integrated view of the information sources created, for which an ontology-based approach seems to be suitable.

3 Mathematical modelling as a knowledge – intensive task

Mathematical modelling of physico-chemical and biological systems is a basic activity in the engineering disciplines, as well as the applied sciences in general. Mathematical models play an important role in engineering tasks like design, control and optimisation as well as in the development of elementary process understanding. The development of mathematical models can in many respects be characterised as a ‘knowledge-intensive’ activity. It involves methodical knowledge at different levels (corresponding to the different modelling steps as discussed later in this section) as well as experience in the underlying domain.

Top and Akkermans described the modelling process as a design-like task, implying a creative, iterative activity accompanied by a learning process [29]. Based on their decomposition of tasks in component-based modelling as shown in Figure 1, they clearly point out that modelling as a creative task starts with specification of requirements. On the background of their evolutionary modelling approach, Top and Akkermans emphasise the revision of initial modelling assumptions. The role of these assumptions has to be considered in order to make the conditions explicit under which a model can be validated. Their analysis of the modelling process shows, that from the documentation point of view not only the result but also the process itself needs to be considered in order to support reuse of models at the knowledge level.

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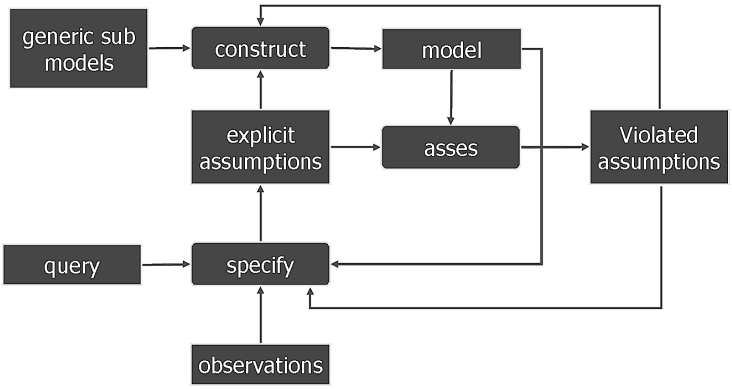
Figure 1 The modelling-process for component models as characterised by Top and Akkermans [29]

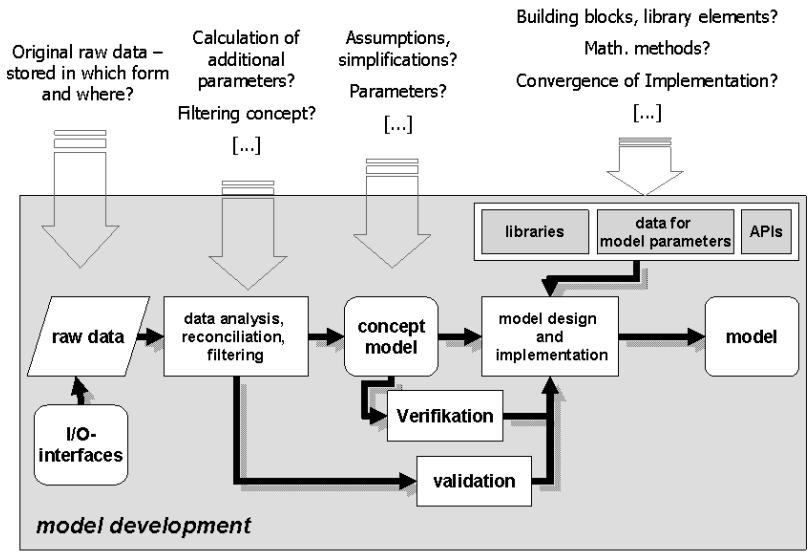


Source: Adapted from Top and Akkermans [29]

3.1 Generated information sources

Figure 2 illustrates the possible information loss connected with model development. The iterative character of the modelling process is neglected for the sake of simplicity. The information lost is required in order to comprehend how data was obtained (measuring principle, calibration information, etc.), how it was filtered and processed (indirect measures, etc.) and in which way it was used for the determination of model parameters as well as for model validation. It also involves practical information concerning the implementation of the model.

Figure 2 Possible 'information loss' during model development



In cases where the particular information is basically available (e.g. a description of the experimental determination of model parameters), it can only accessed 'manually', by going through the available text sources. This prevents advanced queries in order to answer questions such as 'where is the data used to determine parameter *x*'.

The generated information sources are typically heterogeneous in type: measured data (e.g. in form of ascii-files), processed/filtered data (e.g. in form of spreadsheets) as well as reports and publications (including non-textual elements such as equations and figures). Typically, processed data contains metainformation in a non-standardised way, e.g. in a spreadsheet with experimental parameters that refer to columns of measured data and is arbitrarily stored in some extra cells.

3.2 Tool support for the modelling process

Apart from generic modelling environments for the implementation of mathematical models, there are several approaches to support the modelling process in different forms: guidance of the model builder, extended support for reusable components and documentation facilities. Examples are the MODKIT [30] system as a complete modelling platform for chemical processes supporting workflow and model reuse or the concept of Gruber [31] for intelligent assistance of model builders.

This concept in the next section takes a different approach. It does not provide direct support for the modelling process. It targets scientific communities, in order to increase the value of scientific information resources related to mathematical models and supports interdisciplinary communication of the latter at a conceptual level. The concept goes beyond the specification of a fixed set of metainformation for mathematical models (cp. [32]).

4 Ontology-based documentation of mathematical models and related resources

The concept presented in this section has been developed as a base for an information system capable of managing scientific information sources which is integrated in a web-based framework. The latter has been developed for a multidisciplinary research programme dealing with the treatment of industrial waste water.

This section focuses on the use of ontologies, while other aspects of information management for scientific resources, such as requirements for long-term archiving, are beyond the scope of this paper.

4.1 Basic concept

As already mentioned several systems provide direct support for the modelling process, including documentation facilities. The design of those tools is inevitably driven by the requirements of a particular domain. In addition the supporting function of those tools starts at a stage of a modelling process, where several decisions influencing that process have already been made (e.g. at the composition of generic models, cp. Figure 1). While this might be perfectly suitable for the development of models within a domain and with a particular goal (such as a particular engineering activity), we focus on the interdisciplinary development and communication of models. The latter aspect has become more important

with the increasing popularity of the internet as a platform for professional and scientific applications focusing on information exchange.

The concept presented in this section is based on the SchoolOnto approach developed by Buckingham Shum *et al*. [33]. The SchoolOnto-system is a digital library server that has built a methodology of ontology-based document enrichment. It allows one to explicitly represent semantic connections existing within a collection of scientific documents. Connections are based on a discourse-orientated ontology. The design of the latter follows the principle of 'minimal ontological commitment' (see above and [15]). The ontology provides a basic set of possible relations for scientific documents, which is discipline-independent (such as a connection for one approach extending another). In addition, the development of SchoolOnto incorporates a method of incremental formalisation. Documents are enriched with additional information, but the original text remains unchanged.

The design principles of SchoolOnto reflect the fact that a detailed, fixed definition of relations across different scientific disciplines is not suitable. We have developed a concept for the documentation of mathematical models in the area of physico-chemical processes. While our approach allows the representation of more detailed relations between resources than the ones defined in SchoolOnto, we followed the considerations discussed earlier in this paper using the SchoolOnto approach as a starting point. A basic relation (such as the 'extension–relation' mentioned above) is specified with the help of a domain ontology for mathematical models (e.g. by specifying differences in the assumptions used). Using such a top-down approach ensures that a minimum of information concerning relations of different resources is independent of particular disciplines. This maintains the possibility of performing semantic queries without necessarily committing to the domain ontologies used for the specification of detailed relations.

The concept described above has been developed to complement a concept for the standardisation and modularisation of text-based information resources based on XML (cp. [4]). This way it is possible to utilise the occurrence of standard elements in scientific applications, such as parameter lists. The framework presented here allows a mapping of such standard elements to a domain ontology for domain-specific mathematical modelling. These mappings can be defined to be required in order to ensure a minimum of metainformation. Based on the standardised document structure and named elements in the ontology the system is capable of generating proposals for such mappings. This concept is even more important for non-textual sources, such as measured and calculated data. Apart from the standardised structure, a mapping for parameters and physical units to the domain ontology is always required before a particular data set can be archived.

The focus of this concept, concerning the methodology of integration of different information sources, lies in the provision of a common user interface, allowing us to comprehend the development of particular mathematical models and retrieve and access associated information sources. That user interface needs to be adjustable at run-time, since only a small top-level ontology is unchanging, while domain-specific ontologies can be complemented and extended.

It is important to mention, that the concept described here does not (yet) target distributed sources but the centralised storage of standardised information sources. That is why a complex wrapper-mediator architecture is not needed at this point. In addition does the concept takes advantage of the fact that the information sources considered here are in most cases immutable, manifesting a particular state within a research project.

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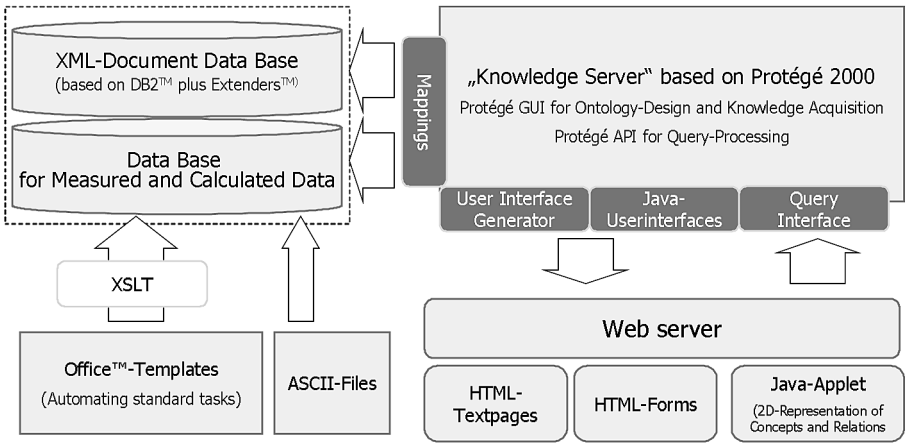
The management of some resources, such as publications, involves legal aspects, which need to be clarified with the copyright holder. A discussion of such aspects is not within the scope of this paper.

4.2 Implementation

The basic architecture of the implementation is shown in Figure 3. The framework includes templates for the creation of standardised documents (including conversion capabilities for existing documents), a database for the storage indexing of those documents, a database for literature, a database for measured and calculated data (which has not yet been implemented) and an ontology server with an editor. The latter is based on the open-source tool Protégé 2000 [34]. This tool allows the creation of ontologies (including several features such as multiple inheritance, and class-instance relations on two levels) and automatically builds user interfaces. The software package Protégé 2000 has been developed at the Stanford University for the creation of ontologies and knowledge acquisition tools. For an evaluation of the predecessor version Protégé Win refer to the publication of Duineveld *et al*. [35].

Figure 3 Basic system architecture



The current version of Protégé 2000 is Java-based, supports the RDF-format and offers a variety of plug-ins which extend the basic functionality. For the framework described here, several plug-ins have been developed for the user-interfaces of Protégé 2000, such as a plug-in for mathematical formulas (see Figure 4), a plug-in for the (reading) access to structured documents and a plug-in for the graphical representation of relations of concepts (currently under development).

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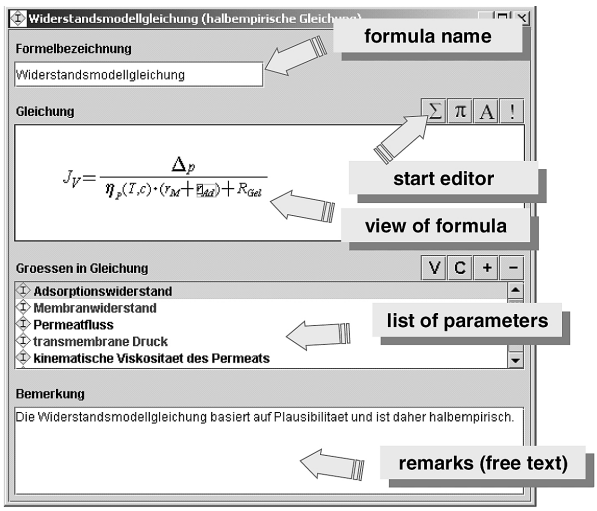
Figure 4 Acquisition of information about model equations and parameters in Protégé 2000



Queries can be submitted via HTML-forms or with help of a Java-applet on the basis of the graphical representation mentioned above. Results are returned in HTML-format and also via applets. This functionality is realised by a servlet layer directly communicating with the Protégé API. Query results are provided in XML and returned as HTML via XSL-processor.

The forms generated by Protégé 2000 serve as a base for the integrated management of different resources. Structured text sources in the form of standardised XML-documents are 'translated' into Protégé 2000 objects using a compositional data model ('document consists of chapters; chapters consist of . . .'). This allows for the creation of semantic connections of documents or parts of a document to other sources ('chapter . . . refers to . . .'). Structured sources in form of databases are mapped to Protégé classes, which makes it possible to view the data stored as Protégé Objects and also supports semantic connections. In both cases, only a reading access is provided concerning the sources themselves.

Several ontologies have been constructed that describe mathematical models and related concepts in the domain of waste water treatment such as an ontology for biological, chemical and physical treatment processes.

The framework is implemented in terms of a prototype system, which provides a basic functionality for the documentation and interdisciplinary communication of mathematical models on the basis of a semantic information concept. The top-down approach described and the ability to create user-interfaces at runtime ensures a minimum of flexibility.

5 Implications for SMEs and industrial enterprises

Regarding research institutions, the benefit of the approach presented in this paper is quite obvious. Concerning the results of fundamental research, scientific institutions are bound to the principle of sustainability (often as a requirement for funding). The approach

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described permits the well-organised use of existing scientific work in several ways. Scientists are enabled to retrieve, access and classify resources efficiently, which helps to build upon existing work and to avoid redundant work. In addition, this approach enhances information sharing among different scientific disciplines thus supporting transparency and quality assurance.

Apart from the latter aspect, the relevance of the advantages mentioned above is not so obvious concerning SMEs and industrial enterprises, since information management causes extra costs (for implementation and maintenance). Once there is no superordinate interest in the long-term availability of certain information resources, that extra cost must be justified with an extra value. At this point it must be considered that information management needs to be well directed in order to be successful. At the same time there is a chance to overcome the problems inherent in many data-centric approaches. Let us consider professional Laboratory Information Management Systems (LIMS) for a brief discussion of this point. Stafford [36] states that typical LIMS lack flexibility and do not provide sufficient information about *processes* but simply their in- and output together with extra data necessary for regulatory purposes. This often resulted in the frustration of users and a very limited acceptance of certain tools. One of the lessons learned is that the focus should not simply be on the kind and amount of data to be stored but also on the infrastructure required to provide *valuable* information.

The approach described in this paper shows that it is possible to separate data and useful knowledge about that data in a flexible manner. Therefore IT-projects in the fields of physico-chemical processes should consider the extra investment in intelligent information integration in addition to existing software instead of targeting at inflexible 'one-does-it-all'-tools.

When combined with product documentation, advanced information management can offer extra sales arguments. This holds for 'knowledge-intensive' products which require a high level of user qualification (such as simulation and synthesis tools). An interactive user-interface offering simple as well as sophisticated queries together with the visualisation of complex correlations is more attractive (and useful) than a voluminous text-based manual.

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